Date :.....

x itr $\Rightarrow \frac{n}{2^{x-1}} \Rightarrow 1$ (a single element lasts in the end).

Solve:

$$n = 2^{x-1}$$

$$\log_2 n = \log_2 2^{x-1}$$

$$\log_2 n = x - 1$$

$$x = \log_2 n + 1$$

$$\therefore x = \log_2 n$$

We ignore const in Time Complexity

$$\therefore x = \log n$$

Number of iterations

So, for input size of n, $\log n$ iterations will be needed in binary search.

$\therefore$ Time Complexity : $O(\log n)$.